161.304: Statistical Modelling

Michelle Marraffini[1] and Matthieu Vignes[2]

School of Mathematical and Computational Sciences
Massey University

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[1] m.marraffini@massey.ac.nz
[2] m.vignes@massey.ac.nz

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Lecture 1 : Paper Overview

- This lecture provides an overview of 161.304.
- Touches on content, organization, and resources.

Paper Coverage 1/2

1. **Statistical models**
   - Quantitative scientific thinking into statistical modelling. Using R.
   - Bayesian data analysis: counting in probability, prior, likelihood, posterior. Grid approximation.
   - Making predictions and working with the posterior.
2. **Revisiting linear regression**
   - In a Bayesian framework. Quadratic approximation of the posterior. Visualisation of prior/posterior.
   - Polynomial and spline regressions
   - Multiple regression and...
   - Causal considerations: DAGs, confounding, biases, etc.
   - GLM, a light introduction

Paper Coverage 2/2

3. **Model choice**
   - Overfitting, regularisation and out-of-sample performance assessment
   - Interactions and "triptych" plots
   - Entropy
4. **Approximation by simulations**
   - Markov chains
   - MCMC methods

Computing

- The paper 161.304 is about thinking and computational methods for statistical modelling and inference.
- Computing will be done in the software R, using the Rstudio environment and with a strong reliance on the `rethinking` package. Make sure you install it asap.
- Computer practicals are an essential element of 161.304: get your hands dirty! Each week.

Assessment

- Two assignments during semester, each carrying weight 25%. One test worth 6%
- Assignments and test require submission of electronic files of results on Stream
- Assignment deadlines:
  1. Friday 16 August 2024 (assign. 1)
  2. Friday 20 September 2024 (test)
  3. Friday 11 October 2024 (assign. 2)
- If you need a bit more time to submit, you **MUST** contact us **before** the deadline. Typically, a one-week extension is the maximum we can grant you. Penalty may be applied.
- The final project (no exam) carries weight 44%; deadline: Friday 8 November 2024; see the page on Stream

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161.304 Stream site

Check the Stream site for 161.304 on a regular basis for:

- Announcements and news
- Study materials in electronic format
- Assignments in electronic format
- Data files for download.

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Become a class rep

- We are recruiting students to volunteer as class reps
- See the Stream website for all the information
- Information video and Even more information



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Any Questions?

